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Editorial Office
Communications AI & Computing, Nature Portfolio

Dear Editor,

We are pleased to submit our manuscript, "Accelerating physics-informed neural networks for full waveform inversion using a hybrid quantum-classical finite-basis architecture," for consideration in *Communications AI & Computing*.

Full waveform inversion (FWI) reconstructs material properties from recorded wave data and underpins applications ranging from seismic imaging to medical ultrasound tomography, yet it remains computationally demanding. Our manuscript presents, to our knowledge, the first integration of parameterized quantum circuits (PQCs) into domain-decomposed physics-informed neural networks (finite-basis PINNs, FBPINNs) for this class of inverse problems. Both the decomposed wavefield network and the global velocity network are realized as classical-to-quantum pipelines terminating in PQCs, implemented as a JAX-native differentiable statevector simulator. This design enables exact, end-to-end automatic differentiation through the classical network, the quantum circuit, and the physics-informed loss, avoiding the overhead of parameter-shift rules.

On a synthetic acoustic FWI benchmark, the hybrid model reaches a lower L1 velocity error than the classical FBPINN baseline and all 15 classical hyperparameter variants in approximately 8× fewer training iterations, while using about 33% fewer trainable parameters. A second benchmark confirms that the same architecture recovers more spatially structured velocity models without additional tuning.

We believe this work fits the scope of *Communications AI & Computing*, in particular its interest in hybrid quantum-classical systems, AI for scientific discovery, and computational modeling in the physical sciences. Because the framework targets the general structure of physics-informed wave inversion rather than any geophysics-specific feature, we expect it to be of interest beyond seismology, including for medical ultrasound tomography, photoacoustic imaging, and non-destructive evaluation.

We confirm that this manuscript has not been published elsewhere and is not under consideration by another journal, and that all authors have approved its submission. We suggest the following reviewers, whose expertise spans physics-informed neural networks, full waveform inversion, and quantum machine learning:

Thank you for considering our submission.

Sincerely,

Hoang Anh Nguyen
on behalf of all co-authors